

⑩ Statement & Course of Action

→ The first step to → solve a problem ^{and} minimize a problem

• The soln shouldn't give other problems.

① S: on a particular highway, no. of road accidents have increased by many folds.

A: Personal monitoring should be done by the police officials on each & every vehicle

Feasible

impractical

② S: Company Z is incurring huge losses due to many competitors in the market.

A: ~~if~~ Company Z should offer a huge discount on all its products & attract customers.

~~if~~ Company Z should study its competitors' products & methods.

→ never choose an action whose outcome we can never predict.

③ S: Concerns have risen about air pollution in Indian cities. A recent report by the World Health Organization states that of the 20 most polluted cities in the world, 13 are in India.

A: ~~if~~ Pollution Committee should immediately take certain measures to control the pollution levels in India.

2) Licenses of polluting industries should be cancelled & owners should be put behind the bars. → Harsh Statements only II follows

④ S: The authorities in society X are blaming them for traffic jams near their society.

A: ~~if~~ street hawkers should not be allowed during peak hours. → difficult to implement

~~if~~ Street hawkers should be warned & asked not to create chaos.

only II follows

⑤ S: many people in city Y are suffering from flu.

~~if~~ the municipal corporation should take immediate steps to control the disease

~~if~~ People in the city should be advised to take certain measures from their end to prevent the disease.

Both I & II follow

⑥ S: According to a survey, the current water reservoir will deplete by 50% by 2050.

A: ~~if~~ Govt of all the countries should appeal to all the citizens to use water carefully & avoid wastage.

X Scientists should start finding an alternative ~~soln to water~~ ^{disaster} ~~practical~~ ^{only I follows}

⑦ S: An ~~earthquake~~ ^{disaster} measuring 3-7 hit Delhi & its surrounding areas a little before 3 Pm today.

A:

X Govt should immediately provide financial assistance to the families.

X People blame the govt for not taking appropriate preventive measures.

Neither I nor 2 follows

⑧ S: The World Bank estimates that India is one of the highest - ~~burning~~ ^{harmful} countries in the world for the no. of children suffering from malnutrition.

A: Govt, People & various NGOs should come together & try to eradicate the problem.

X Govt. should take ^{financial} ~~help~~ ^{doesn't require} from foreign countries to eradicate this problem.
only I follows

⑨ S: many students have gone on a mass ~~have in~~ ^{protest} protest against the institute's new policies.

A: The institute should immediately ~~withdraw~~ ^{discuss} the new policies.

X All these students should be suspended.
Neither I nor II follows

⑩ S: Increase in the pollution levels in the city by way of automobile ~~exhausting~~ ^{exhausting} poses a severe threat to the inhabitants.

A: All the automobile factories in the city should immediately be closed.

X Govt should run programs to promote the use of public transport & car-pooling by commuters.

X The govt should allow any licensed factories & vehicles.

Both II & III follows

⑪ S: During the recent Diwali season, there was a rise in criminal activities.

A: The police should immediately investigate into this.

X The police should take adequate precautions to avoid thefts during Diwali season.

X The known criminals should be arrested before any such season.
Both I & II follows

(12) S: Since America is not likely to grant Pakistan's request for \$300 billion fast disbursing loan for the current year.

A: ~~1~~ Pakistan should ask other countries for the loan.

~~2~~ Pakistan should talk to America to meet the loan giving it persuasive reasons & explaining the urgency.

Both I & II follows

Either I or II follows

(If I is done, no need of II & vice versa)

(13) S: The Japanese petroleum industry venturing into the West European markets faces tough competition from the Chinese.

A: ~~1~~ Japan should search for other markets for its products.

~~2~~ Japan should improve the quality of the petroleum products to compete with the Chinese in capturing these markets.

Running away from market is not an option (But here the case is not that)

Both I & II follows

(14) S: Five terrorists were caught with illegal explosives in the city.

~~1~~ The police should be on high alert to

avoid any untoward incident.

~~2~~ The 5 terrorists should be set free & their movements should be watched to infiltrate the terrorist syndicate.

only I follows

(15) S: Villages in Rajasthan have been severely damaged due to drought. This has caused a financial burden on State Govt.

A: ~~1~~ People of the area should be shifted to other suitable places. ~~2~~ State Govt should ask for some financial support from central Govt.

Both I & II follows (But that's not in

options, $\therefore$ none of these)

(16) S: Not just looks, internal beauty is extremely important.

A: ~~1~~ Pay attention to fashion.

~~2~~ Don't pay attention to fashion.

~~3~~ Books on self-development & spirituality should be encouraged.

looks and temp a little bit

inner beauty - advice

Both I & III follows

(17) Visual / Spatial reasoning

→ Translation

→ Scaling

→ Rotation

→ Mirroring

→ Assembling

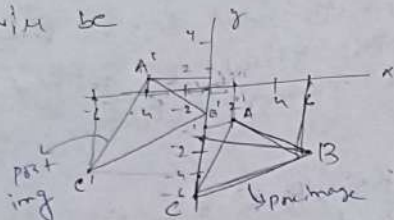
→ Grouping

→ Paper cutting

→ Paper folding

→ Pattern in 2D & 3D

① The translation Rule for the image will be



$$\begin{aligned} A &\rightarrow A' \\ B &\rightarrow B' \\ C &\rightarrow C' \end{aligned}$$

For A take one point, & the translation for that 1 point is applied to all automatically & that's the same img.

$$A(2, -1) \xrightarrow{-6} A'(-4, 1)$$

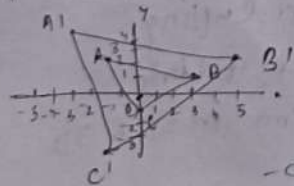
$$\therefore (x-6, y+2)$$

[Check for B & C,

$$B(6, -3) \xrightarrow{-6, +2} B'(0, -1)$$

$$C(0, -6) \xrightarrow{-6, +2} C'(-6, -4)$$

② The Scale factor for the dilation



$$AB \rightarrow A'B'$$

• Apply for 1 line $\rightarrow$ get the scaling factor & automatically if u have scale factors for

other lines too. (As it's the same image).

$$\begin{aligned} AB &\rightarrow A'B' \\ |AB| &\rightarrow |A'B'| \end{aligned}$$

$$A(-1, 2) \quad B(3, 1)$$

$$|AB| = \sqrt{(3+1)^2 + (1-2)^2} = \sqrt{17}$$

$$A'(-3, 4) \quad B'(5, 2)$$

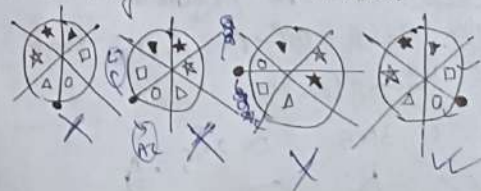
$$\begin{aligned} |A'B'| &= \sqrt{(5+3)^2 + (2-4)^2} \\ &= \sqrt{68} \end{aligned}$$

$$\begin{aligned} \text{Scale factor} &= \frac{\text{Post img}}{\text{Pre img}} \\ &= \frac{\sqrt{68}}{\sqrt{17}} \\ &= 2 \end{aligned}$$

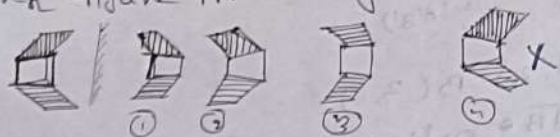
③



Which fig is a rotation of the object?



④ choose the correct mirror image of the given figure from among 5 the four alternatives



original mirror
L → R
R → L

TOP & right
bottom as same

⑤ choose alternative which is closely resembles the water image of the given combination.

BK 5 O R P 6 2 → water image

① B K 2 O B 6 9 5 X ② B K 2 O B 6 C 5

③ B K 2 O B 6 C 5 X ④ B K 5 O R P 6 C 5

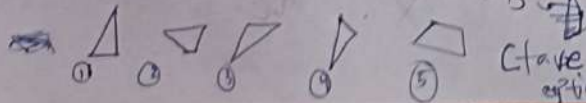
watering
top → Bottom
Bottom TOP

L & R which
same

→ B K 2 O B 6 C 5

⑥ select the alternative which helps three out of the 5 alternative figures which when fitted into each other would form an equilateral triangle.

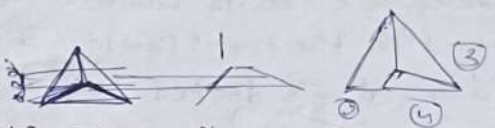
A. 1 2 3 4 B. 2 3 4 C. 3 4 5 D. 2 4 5



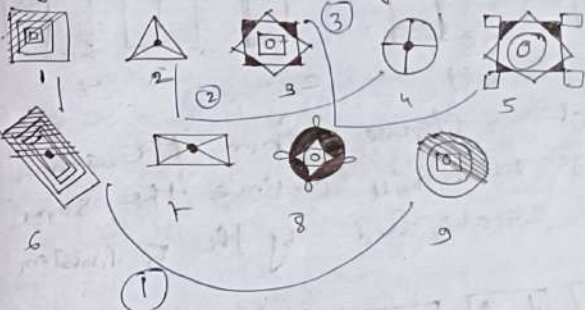
Give the help of options

1 2 3 →

3 4 5 →



⑦ Group the given figures into 3 classes using each figure only once.

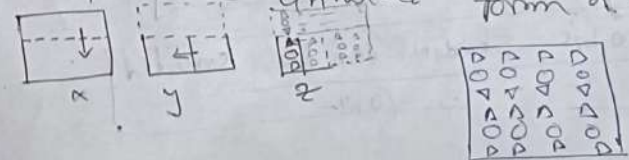


1, 6, 9 → ①

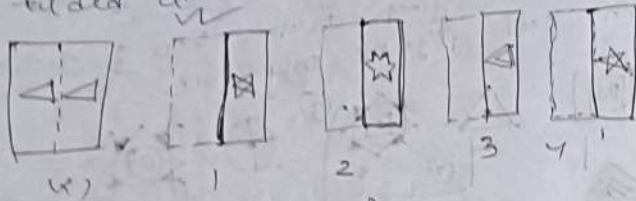
2, 7, 4 → ②

3, 8, 5 → ③

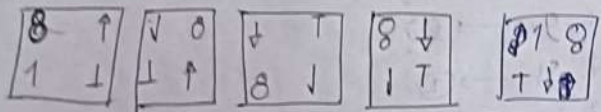
⑧ Question consists of a set of 3 figs x, y, z showing a sequence of folding of a piece of paper. Fig (z) shows the manner in which the folded paper has been cut. These three figs are followed by four answer figs from which you have to choose a fig which would most closely resemble the unfolded form of figure (z).



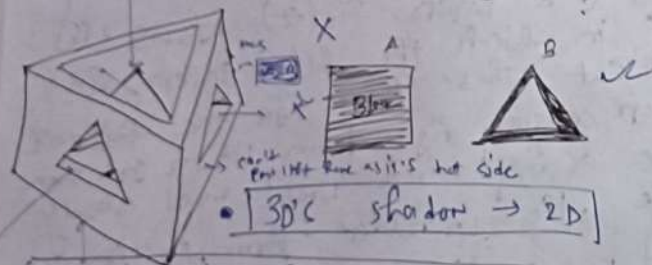
⑨ Pick out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is held at the dotted line.



⑩ Select a figure from the four and figs which will continue the same series as established by the 5. Problem figs



⑪ Which of the shadows on the right can result from casting light on one of the 3D shape's sides?



• closed obj's shadow can be created
• opened

⑩ Seating Arrangement (Dates arrangement)

• Tips

Steps

i) Count

ii) Draw frame

iii) Solve sentence by sentence & then merge them

① A, B, C, D, E, F & G are 7 friends sitting in a single row facing North.

i) D is to the immediate right of C.

ii) E & A are neighbours of F.

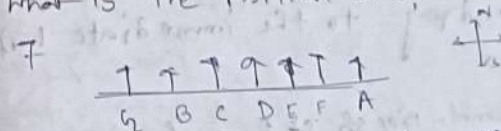
iii) B is to the immediate left of C & on second place from left most end.

iv) A is at the right most end.

a) Who is in HW C & E?

b) Who are the neighbors of A?

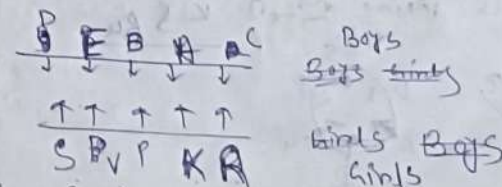
c) What is the position of E?



a) D, b) B, c) From left - 3rd
from right - 3rd

② [Not mixing the 1st of arrangement]

A, D, E, C, B | P, K, R, S, V



Who is in the middle of the row of boys?
Who is immediately to right to Pankaj?

- Q Who is opposite to of Dipesh?
 d) How many boys are there between Ashwin & Bipin?

a) E, b) K, c) S & D

③ $\begin{array}{c} m \text{ } g \text{ } p \text{ } a \text{ } s \text{ } k \\ \downarrow \downarrow \downarrow \downarrow \downarrow \\ \text{TTTTT} \\ \text{D } k \text{ } R \text{ } A \text{ } B \text{ } P \end{array}$ (G)
 (B)

- a) which girl is facing Rupesh?
 b) which pair of girl & a boy is at one of the extreme ends?

- Komal - Dipesh
- meena - Kamlesh
- Geeta - Suresh
- Geeta - Parash

c) who is standing to the immediate left of Rupesh?

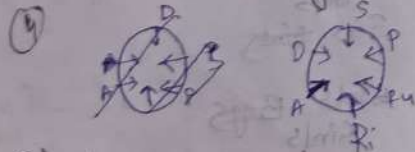
a) Radha, b) Komal - Dipesh c) Ramesh

d) which option is not same as others?

i) Ramesh - Radha, ii) Dipesh - Geeta

iii) Suresh - Komal, iv) Rupesh - Radha

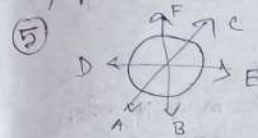
c) who are immediate neighbours of the one who is facing Rita? K & Bu



a) who are the neighbours of Dipen? S, A

b) who is sitting opposite to Dipen? Ru

c) who is just right to Ashwin? Ru

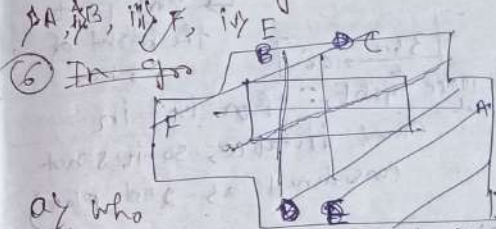


a) who is b/w D & B? A

b) who are the neighbours of A? D & B

c) who is exactly to the right of D?

PA, PB, PC, PD, PE



a) who

is to immediate right of A? E

b) who is b/w B & F? C

⑦

if nothing is said by default - Inward Slating

$N(m) R(h) T$

$(f) L \leftarrow \rightarrow m(f)$

$P \downarrow S \downarrow O$

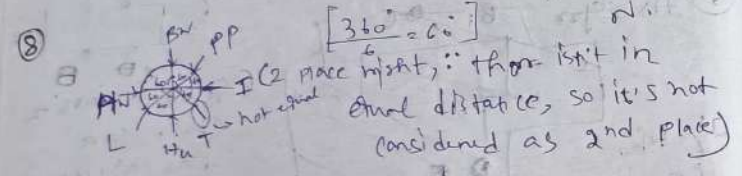
a) which of the following is true with r. t. given seating arrangement?

i) No two males are immediate neighbours of C/O. $\rightarrow F$

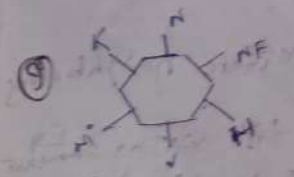
ii) R & S face C/O others in the seating arrangement $\rightarrow F$

iii) L sits in the center of one of the sides of the square table. $\rightarrow T$

- Q. 9 is a male & sits diagonally opposite to P → F
- Q. by who is m's husband? → P
- Q. How many people sit b/w m & N when counted in anticlockwise direction from m? → 2
- Q. who is the wife of N? → 0
- Q. what is the position of P w.r.t. N? → 2nd pos to the right of N



- Q. who is the only person sitting on the immediate left of BW? → PP
- Q. i. Lavi is on the right of Hulk. ii. HW is sitting next to Shannon. iii. There is sitting adjacent to Lavi.
- Q. the angle b/w third Hulk in the anticlockwise direction? → 330°
- Q. where is HW sitting with r. t. BW? → this immediate right
- Q. There is sitting b/w which pair? → H & I

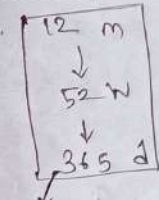


- Q. who is placed exactly opposite to N? → H
- Q. If one neighbour of N is of V, who is the other? → K
- Q. first sequence - N, NF, H
- Q. who is at the same distance from V as K is from V? → NF
- Q. Neighbour of H - NF, V

11) Calendars

Concepts

Normal years

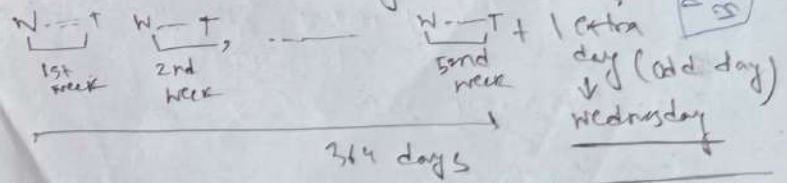


How many extra days are remaining? → $\frac{365}{7}$ (extra day) = 52 W & 1 day

eg 9n

odd day

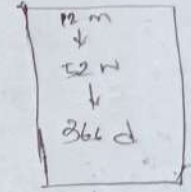
Ex. A year begins on 1st Jan falls on Ned. What will be the day of 31st Dec?



direction blood relation

For a normal year + leap begins on Sat & ends on the same day.

Leap year



Extra Odd days = $\frac{366}{7}$ = 52 W & 1 day

odd day

Wednesday

- For leap years

Wed - T, Wed - T, ... Wed - T + 1 + 1
 1 2 12th 2 days
 ∴ For leap years a week will end on the next day of the start day.
 i.e. if Start → Mon, end → Tue
 " → Tue, end → Wed

So on

- If a year (that can be like this 1986, 1952 - etc/ 1900, 1600 like this (century years)) is divisible by 4 or 400 [For year like this 1956, 1982 → take last digit & divide them by 4, if completely divisible then → leap years]
 For century years → Divide by 400 → If completely divisible then → leap years

- For Century years

For 2 leap years, there falls 3 non-leap years
 Century years

Q. Does a leap year come after every 4 years?

No, eg → 1896 - 1900 - 1904

but why so,

Till now we know → 365 d + (6 hrs) → 1y

100, 1800, 2000
 century + 4
 So this makes diff. over 100 years
 5 hrs 48 min → 6 hrs

How many leap years in 100 years?

No. of years	Leap years	Normal years
100	24	76
200	48	152
300	72	228
400	97	303

$$(1-100)/4 = 25 \text{ ly} - 1 = 24 \text{ ly}$$

$$75 \text{ ly} + 1 = 76 \text{ ly}$$

century not

$$\frac{100}{400} = 0.25$$

∴ It's not leap but normal years

- odd days

No. of days Day of the week

0 → Sun
 1 → Mon
 2 → Tue
 3 → Wed
 4 → Thurs
 5 → Fri
 6 → Sat

10 days → 1w + 3 days

Jan (3) d → 4w + 3 days

365 d → 52w + 1 day

- No. of odd days in a century

[illegible]

- when it's a leap Century Year (1800 - etc)
then we'll have 0 or 0 of odd days

Year	No. of old days
1600	0
2000	0
2400	0

① What was the day on 1st April 1896?
6 apr. 1896 (Friday)

6 apr. 1856 (Jung m)

1895 is completed

1895 y
1600 y + 2007 + 55 y
↓ 1 ↓ 55 y
0 + 3 23 72
odd days
↓ ↓
23 x 2 1 x 2
23 + 2
= 0 + 3 + 46 + 72
2
121 days
17 + 172
2 days

1895 is completed 8 days
Jan '26 + Feb '26 + Mar '26 + Apr '26
31 days + 29 days + 31 + 6
(∵ 01/4 = 0)
= 97
7
2 13 N + 6 days
(mod 7)

rem od 2 days

$2 + 5$

8/

7
2 1st + 1 day

1 odd day means $\rightarrow$ monday

② What was the day on 25th June 1983?

25th June 1983

1982 has finished

$160 + 300 + 82$
 $\downarrow$
 $0 \quad 1 \quad / \quad 18/21$
 $20 \quad 60$
 $41 \quad \text{my}$
 1
 20×2^{12}

$\therefore 0 + 1 + 40 + 62$
 102

$\text{Jan '83, Feb '83, Mar '83, Apr '83, May '83}$
 $1 \quad 1 \quad 1 \quad / \quad 25 \text{ days or}$
 $= 31 + 28 + 31 + 30 \quad \text{June '83}$
 (NY)
 $+ 31 + 25$
 $= \frac{176}{7}$
 $= 25 \text{ ft } 7 \text{ d}$

Handwritten musical notation on a five-line staff. The notation includes a treble clef, a key signature of one sharp (F#), and a time signature of 6/8. The melody consists of a quarter note on G4, followed by a quarter note on A4, and a quarter note on B4. Below the staff, the text "Handwritten musical notation" is written.

③ Jan 1, 2001 was Monday, that was the day on Dec 31, 2002.

Jan 1, 2001 $\rightarrow$ mon
 LSA is a normal year so, it starts &
 ends on the same day

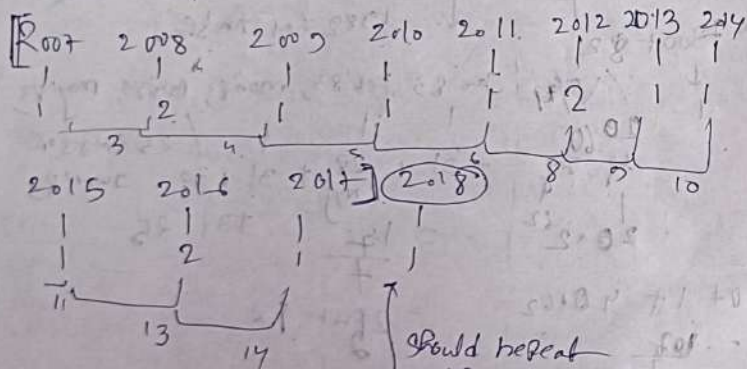
$\therefore 31^{\text{st}} \text{ Dec } 2001 \rightarrow \text{Mon}$
 $\therefore 1^{\text{st}} \text{ Jan } 2002 \rightarrow \text{Tue} \quad \therefore 31^{\text{st}} \text{ Dec } 2002 \rightarrow \underline{\text{Tue}}$

Calendars for 2017 will be same as
calendars for which nearest year in
future? • a calendar repeats in 400y

2007 2008 2009

when the no. of odd days
become 0 the calendar
repeats.

So here
should
be 2407
(but it is
far, not
near)



14 - 7 = 7 $\therefore$ 0 odd days

$\therefore$ 2017 (keep in mind if the year is
leap, then the same / or vice versa
nearest year will also be
leap, if not then not, otherwise
for odd days, it will make
diff.)

Q5 Which day can't be the last day of the
century?

From the table we can see century years
have specific no. of odd days.

year	odd days
100	5
200	3
300	1
400	0

after
this the
table will
repeat

& we also know these odd days has corresponding
days,

5	→	the last day
3	→	Fri
1	→	Tues
0	→	Mon

the, thurs, sat
don't make the
table.

So, they can't
be the last
day of century.

(the last day
of century will
be in the these
days)

Q6 How many days are there in 'y' weeks, 'y'
days?

$$\begin{aligned} 1N &= 7d \\ yN &= 7y d \\ + & \\ y \text{ days} &= y \therefore 7y + y = 8y \text{ days} \\ \therefore & 8y \text{ days} \end{aligned}$$

Q7 15th Jan 1977 was a Sat. what was the
of the week on 15th Jan 1976?

$$\begin{aligned} 15 \text{th Jan } 1977 & \text{ (Sat)} \\ & \downarrow 6 \text{ odd days} \\ \therefore 6 - 2 &= 4 \rightarrow \text{Thursday} \end{aligned}$$

15th Jan 1976
 $\therefore$ i.e. it has
2 odd days

⑧ Calendar for the month of Apr looks same as that for month of _____

(carry over days over)

After 0 odd days calendar begins to repeat.

Apr May June July
30 + 31 + 30 + 31

31
27

= 13 not 0 odd days

after that the calendar will repeat.

July month will be same as April.

⑨ What was the day of the week on 5th Apr 1995?

1794 5th Apr 1995
1600 + 100 + 94 + Jan + Feb + Mar + 5 days of Apr
= 6 + 5 + 23 + 31 + 28 + 31 + 5
= 223 + 71
= 294

217
7

= 31 + 0 odd days

∴ Sunday

15th March in given year was Sunday. What day was it on 1st Nov?

15th March 21 years (rotating given by default)
18 + 30 + 31 + 30 + 31 + 30 + 31 + 30 + 31
= 244/7
= 34 not 6 days

Sat

Clocks (1-2 hrs)

TIPS

What will be the time in mirrors?

What will be the time in the original clock?

11:60 - whatever the time it is in actual/original clock

5 spaces
1 hr → 5 spaces
12 " → 60 "

these spaces are called minute

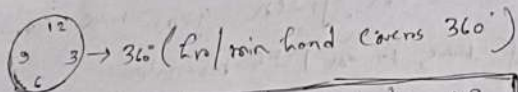
from 12 am - 12 pm → There are 60 min spaces in a clock

min hand moves 1 min → That means min hand moves 1 min spaces
2 min → 2 min spaces

like this

60 min →

60 min on 1 hr



In 1 hr min hand goes 360° in 1 hr

→ this is how much deg. (i.e. in 1 hr the min hand moves how much deg.)

$$\begin{array}{l} 12 \text{ hrs} \rightarrow 360^\circ \\ 1 \text{ hr} \rightarrow \frac{360}{12} \\ = 30^\circ \end{array}$$

In 1 hr hr hand goes 30° min hand goes 360°

In 1 min second hand goes 360°

How much does the min hand goes

$$\begin{array}{l} 60 \text{ min} \rightarrow 360^\circ \\ 1 \text{ hr} \rightarrow 6^\circ \end{array}$$

In 1 min min hand goes 6°

1 hr $\xrightarrow{\text{hr hand}}$ 30°

60 min $\rightarrow 360^\circ$

$$\begin{array}{l} 1 \text{ hr} \rightarrow \frac{360}{60} \\ = 0.5^\circ \end{array}$$

In 1 hr hr hand moves 0.5°

1 hr $\xrightarrow{\text{min hand}}$ 60 min space

60 min $\rightarrow 360^\circ$

1 hr hr hand 5 min " the min hand has gone 60-5 = 55 min ahead.

In 1 hr the min hand goes/gains 55 min space

to the hr hand



Whenever it's 3, 6, 9 etc → The angle

15:30

15 min spaces are there

Whenever he want an angle of 90° , the min & hr hands will be 15 spaces apart.

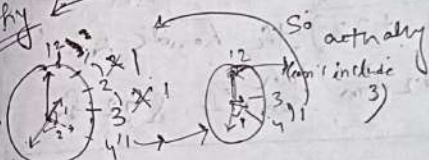
min 90° angle

Every 1 hr we can find this position, $\frac{360}{90} = 4$ times

For 12 hrs $\rightarrow 4 \times 12 = 48$ times

1 day $\rightarrow 48 \times 2 = 96$ times

Why



∴ total 3

same for

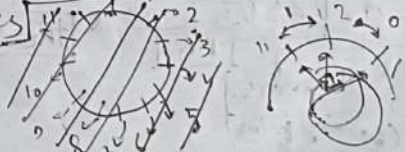
∴ $3 \times 3 = 9$

For every 1 hr, the min & hr hand coincide

for one time

For 12 hrs it should be 12 times but, no

it is 11 times



1 coincidence only

Coinciding $\rightarrow 0^\circ$, opposite to 180°

• In 1 hr, hr & min hand marks (together) 180°
 [for 1 time.] $180^\circ = 180^\circ = 11 \text{ times}$

∴ In 12 hr ————— = 22 times

∴ In 1 day there will be 22 times coinciding & 22 times facing opposite of min & hr hand.

• Both hr coincide & facing opposite, the min & hr hands are in straight line (not hr & min etc. they are in straight line). In 1 day for 44 times (22 + 22), the hands will be in straight line (22 for both coincide & 22 for opposite facing).

• If 1 clock is showing 12:10 pm, that's the correct time & the other clock is showing 12:15 pm, that means the 2nd clock (12:15) has gained 5 more mins. / the other 2nd clock is ahead by 5 mins.

• Similarly, if 12:05 pm → the clock is behind by 5 mins.

• Fast clock (gaining mins) → A clock that gains 2 min / 2 sec or can say 2 unit of time each hour. eg = C.T → 12:00 → 1:00
 but 1:02 → 2:00

(should be)

but 2:04 → (losing mins)

• slow clock → The clock that loses 2 unit of time per hr.

eg - C.T → 12:00 → 1:00
 but 12:55 → 1:50 (not 2)
 (let say losing 1 min / hr)

• How we express time

i) Past 15 → 15 Past 7 → 7:15

ii) 20 mins to 8 → 7:40

iii) Quarters Past 6 → 6:15

(15 min)

iv) Quarters to 6 → 5:45

v) Half Past 6 → 6:30

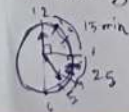
vi) midday → 12:00 pm, mid night → 00:00 or 12:00 am

vii) Am → 00:00 am → 12:00 pm
 Pm → 12:00 pm → 00:00 am

• A broken clock gives correct time twice per day. ↓ exceptional (might be)

• For slow/fast clock it can be 2 or 1 or not correct at all, i.e. it might be, there is so such guarantee.

① What angle is formed b/w the min & sec hands of clock if they are 25 mins apart.

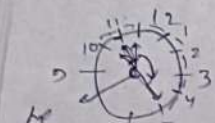
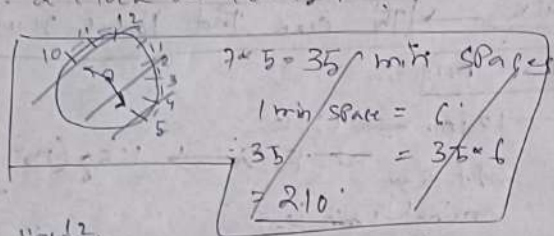


15 → 90
 10 → 15
 60 → 15
 15 → 60

$$90 + 60 = 150^\circ$$

OP 6 min sp → 360
 1 → 360
 25 → 6 × 25 = 150

2) What will be angle b/w the two hands of a clock at 10:25 PM?



He is calculating this angle (1st x)

$10:25 = 60 + 150 = 210^\circ$ by min hand
 In 1 hr, hr hand covers 30°

But when the min hand moves the hr hand also moves correspondingly.

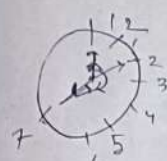
∴ the angle that has moved with min hand correspondingly.

1 hr hr hand moves 30°
 60 min $\rightarrow 30^\circ$
 $1 \rightarrow \frac{1}{2}$
 $85 \rightarrow \frac{1}{2} \times 85$

∴ the total angle turned by the movement of hr hand $30 + 12.5$

312.5°
 $x = 312.5 - 150 = 162.5^\circ$

3) A clock is started at noon. By 1.0 mins past 7, the hour hand has turned through —



hr hand
 $1 \text{ hr} \rightarrow 30^\circ$
 $7 \text{ hr} = 210^\circ$

When the min hand moves 1 min then also hr hand will make some angle.

$1 \text{ min} \rightarrow \frac{1}{2}^\circ$
 $10 \rightarrow 5^\circ$

∴ total deg. the hr hand turned = 215°

4) If the min hand of a clock has moved 300° , how many degs. has the hr hand moved?



In 1 min $\rightarrow 6^\circ$ [60 min $\rightarrow 360^\circ$]
 $\therefore 6^\circ \rightarrow 1 \text{ min}$

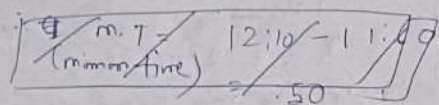
$300^\circ \rightarrow \frac{300}{6} = 50 \text{ mins}$ have been passed
 Now, In 1 min hr hand moves 0.5°

$50 \rightarrow \frac{1}{2} \times 50 = 25^\circ$

5) A clock when seen in a mirror shows 11:25. What is the correct time?

$C.T = 11:60 - 11:25$
 $= 00:35$ or $12:35 \text{ (am)}$

⑥ If real time is 12:10 then what will be mirror image of that clock?



11:50

12:10 (but will get -ve if we do so)

Now, 12:10 can be written as 00:10

12 am can be written as 00:00

11:50

00:10

11:50

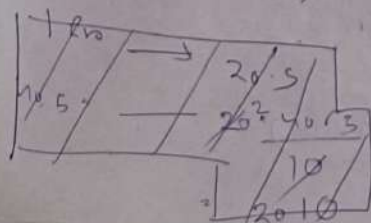
⑦ Ramesh sets his grandfather's clock to the right time at 2 am on Thursday. But he didn't know that the clock gains 20 s for every 3 hrs. What'll the clock indicate on Friday night 6:30 pm?

Starts at 2 am, Thurs

2 am, Fri } 24 hrs

2 pm, Fri } 12 hrs

6:30 pm, Fri } 4:30 hrs



40.5 hrs (30 min diff i.e. 5)

20 s → 2 hrs
? → 40.5

$2 \times 20 = 40$
 $3 \times 5 = 15$

$270 \text{ s} = 4.5 \text{ min}$ the clock has gained $\left(\frac{20 \times 270}{20} = 4.5 \text{ min}\right)$

⑧ A clock loses 5 min every hour & was set right at 11 am on a Monday. When will it show the correct time again?

Broken clock → 2 times

1/2 time → 2, 1, X



If it loses 12 hrs → It'll show the correct time at least 1

Here, the clock has to lose 12 hrs

1 hr → 100 s → 5 min
12 hrs → 12 × 60 = 720 min

$12 \times 60 = 720$

After 144 hrs the clock will show correct time again.

It will lose 12 hrs & show correct time

$12 \times 12 = 144$

Fri, 11:50 Sun, 11:50
Sat, 11:50 Sun, 11 am

Mon 11 am
Tue 11:50
Wed 12:40
Thurs 1:30

Q my watch, which gains uniformly, is 2 mins behind at noon on Sunday, & is 4 min 48 sec ahead at 2 pm on the following Sunday. When was it correct?



total gained = 4 min 48 sec
= 48 sec has been gained by the clock

in 170 hrs the clock has gained 48 sec

It showed correct time after gaining 2 min (∵ it was 2 min behind)

now 2 min = 120 Sec

∴ 170 hrs $\xrightarrow{\text{gained}}$ 48 sec
? $\leftarrow$ 120 "

$$\frac{170 \times 120}{408} = 50 \text{ hrs}$$

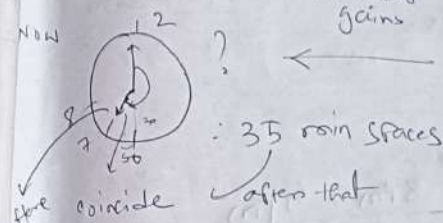
∴ It showed correct time in 50 hrs

48 + 2 = 2 hrs days + 2 hrs

∴ Sun 12 $\rightarrow$ Mon 12 $\rightarrow$ Tue 12 pm + 2 hrs
= Tue, 2 pm

At what time NW 7 o'clock & 8 o'clock will the hands of a clock be together?

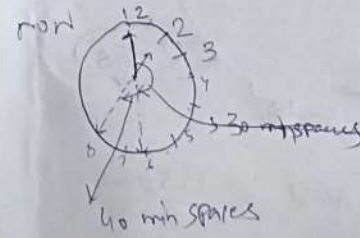
→ Coincide
1 hr $\xrightarrow{\text{min hand}}$ 60 min spaces
1 hr $\xrightarrow{\text{hr hand}}$ 5 "
min hand gains 60 - 5 = 55 min spaces every hr
1 hr: 60 min $\xrightarrow{\text{min hand gains}}$ 55 min spaces



∴ 35 min spaces
after that
 $\frac{60 \times 35}{55} = 38 \frac{2}{11} \text{ min}$, i.e., $38 \frac{2}{11} \text{ min past 7}$

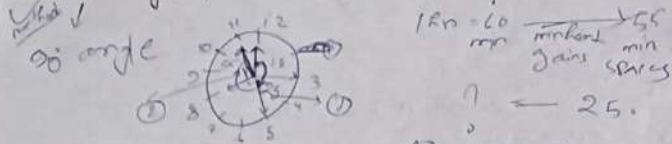
Q When will the clock's hand be in a straight line but pointing in opposite directions b/w 2 am & 3 am?

min hand gains 55 min spaces every hr
1 hr = 60 min $\xrightarrow{\text{min hand gains}}$ 55 min spaces



$$\frac{40 \times 60}{55} = 43 \frac{7}{11} \text{ min past 2 am}$$

(12) When will be both hands of a clock at right angle, b/w 8pm & 9pm?



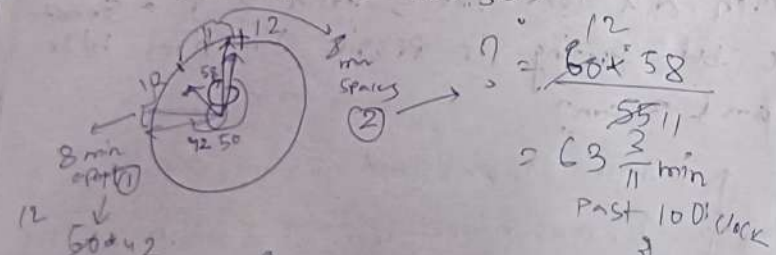
1 hr = 60 min
min hand gains 55 min spaces
? ← 55

$$\text{①} = \frac{55 \times 60}{55} = 27 \frac{3}{11} \text{ mins Past 8 pm}$$

∴ ② = 60 min

8 Past 60 mins Past 8 pm = 9 pm

(13) Ramesh wants to set the clock hands 8 mins apart b/w 10 o'clock & 11 o'clock. What would be the time shown in the clock if he does so?



$$\frac{60 \times 42}{55} = 43 \frac{9}{11} \text{ min}$$

Past 10 o'clock

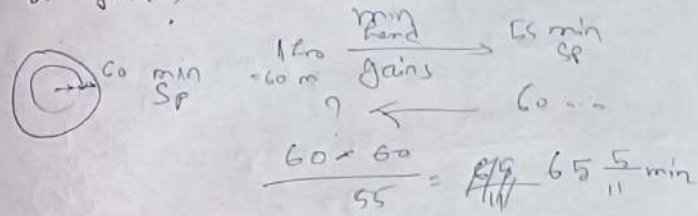
$$\text{②} = \frac{60 \times 58}{55} = 63 \frac{2}{11} \text{ min Past 10 o'clock}$$

2 ans

But as said b/w 10 - 11 so

$$\text{Now, } 63 \frac{2}{11} = 60 + 3 \frac{2}{11} \text{ Past 10} \\ = 1 \text{ hr } + 3 \frac{2}{11} \text{ } \\ = 11 + 3 \frac{2}{11} \text{ o'clock}$$

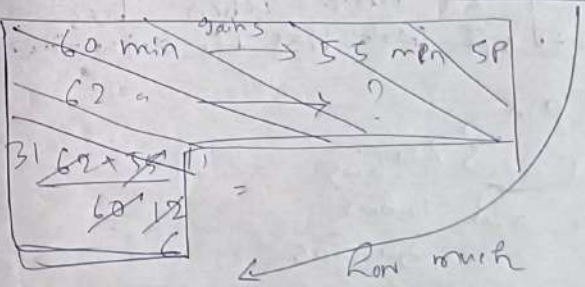
The min hand & hour hand of a clock meet every 62 mins. Does the clock gain or lose in 24 hrs. And how much does it lose or gain?



∴ Every $65 \frac{5}{11}$ min the hr & min hands are together

After $65 \frac{5}{11}$ mins the hr & min hands will always be together.

∴ $62 < 65$
Clock is fast → i.e. its gaining

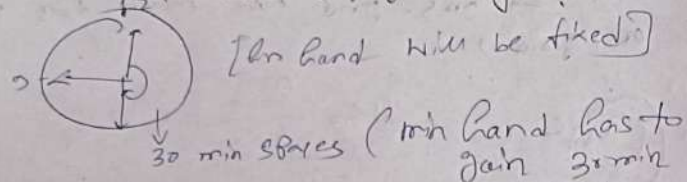


$$65 \frac{5}{11} - 62 = 3 \frac{8}{11} \text{ min Per Circulation}$$

$$62 \text{ min } \xrightarrow{\text{gaining}} 3 \frac{8}{11} \text{ min}$$

$$24 \text{ hrs } \rightarrow \frac{38}{11} \text{ min} \\ \frac{38}{11} \times \frac{24 \times 60}{62} = 78 \frac{14}{11} \text{ min In 1 day = 24 hrs it gains } 78 \frac{14}{11} \text{ mins}$$

- (15) At what time after 9 am, do hands of clock make a right angle?



60 min $\xrightarrow{\text{gain}}$ 55 min sp
 $\uparrow$ $\xleftarrow{\text{SP to get right angle}}$ 30 "

$$\frac{60 \times 30}{55} = 32 \frac{8}{11} \text{ mins Past 9 am}$$

- (16) Jaggu set an alarm on his clock for monday 11 am. He sets his clock correctly on Sunday 1 am. He was unaware the clock gains 20 min every 24 hrs. when his alarm went off what was the actual time?

Sun 1 am
 Mon 1 am } 24 hrs
 Mon 11 am } 10 hrs
 34 hrs

20 m $\rightarrow$ 24 hrs
 $\uparrow$ $\xleftarrow{\text{34 hrs}}$

$$\frac{20 \times 34}{24} = 28.333 \text{ mins}$$

$$24 = 28 \frac{1}{3} \text{ min}$$

= 28 min 20 sec (gained by clock in 34 hrs)

As per clock $\xrightarrow{\text{alarm}}$ 11 am (wrong time)

But As the clock has gained time 50

it will give alarm before 11

$\therefore$ Real time (Real alarm) $<$ 11 am

How much less

$\therefore$ Now the time 28 min 20 sec (that is gained by clock in 34 hrs)

$$\text{Real time} = 11 - 28 \text{ min } 20 \text{ sec}$$

$$10 \text{ am} \leftarrow 60 \text{ min} - 28 \text{ min } 20 \text{ sec}$$

$$10 \text{ am} = 31 \text{ min } 40 \text{ sec Past 10 am}$$